MIL-W-50577C (AR)
20 June 1979
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MIL-W-50577B (PA)
7 February 1976

MILITARY SPECIFICATION

WARHEAD, 155MM, M549
BONDING ASSEMBLY

This Specification is approved for use by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers parts for the 155MM, Warhead bonding assembly, for use in Projectile, 155MM, HE, RA , M549/ M549A1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

UU-T-81	-Tags, Shipping and Stock

MILITARY

MIL-A-48078	-Ammunition, Standard Quality Assurance Provisions, General Specification For

FSC: 1320

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

FED-STD-141	-Paint, Varnish, Lacquer and Related Materials; Method of Inspection and Testing
FED-STD-151	-Metals; Test Methods

MILITARY

MIL-STD-410	-Nondestructive Testing Personnel Qualification and Certification
MIL-STD-1169	-Packaging, Packing, and Marking for Shipment of Inert Ammunition Components

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

PRODUCT AND PACKAGING DRAWINGS

9236005	-155MM Inert RAP Assembly, M549
8837839	-Pallet for Projectile Metal Parts
9235997	-Warhead Bonding Assembly

INSPECTION EQUIPMENT DRAWINGS

9225684	-Recommended Equipment & Procedures for Ultrasonic Inspection
9246484	-Ultrasonic Inspection STD Longitudinal and Radial Cracks M549 WHD Body
9246485	-Ultrasonic Inspection STD Circumferential Cracks M549 WHD Body
9246486	-Ultrasonic Inspection STD (Piping) Eddy Current STD (Nose Cracks) M549 WHD Body

2.2 Other Publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply:

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AMERICAN SOCIETY FOR TESTING AND MATERIALS

E8 - Tension Testing of Metallic Materials
E10 - Brinnel Hardness of Metallic Materials
E18 - Tests for Rockwell Hardness of Metallic Materials

(Copies may be purchased directly from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

ANSI B46.1 - Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, New York 10018.

PANGBORN ROTO BLAST STEEL SHOT AND GRIT CATALOG

(Copies may be obtained from Pangborn Division of the Carborundum Co., Hagerstown, MD 21740).

3. REQUIREMENTS

3.1 Material.- Materials shall be in accordance with applicable drawings and specifications.

3.2 Warhead Bonding Assembly.- The warhead bonding assembly and components shall comply with all requirements specified on Drawing (dwg.) 9235997 and with all the requirements specified in applicable specifications.

3.3 Heat treatment or stress relief. Heat treatment or stress relief applied shall be uniform throughout each lot of warheads and shall be so performed and controlled as to produce in the warhead the mechanical properties specified on the applicable drawing. No warhead shall be reheat treated if it fails to meet the applicable drawing requirements without Government approval.

3.4 Hydrostatic Test.- The warheads shall withstand an internal hydrostatic pressure in accordance with requirements specified on drawing 9235995 without evidence of leakage, cracks, ruptures or permanent distortion when tested as specified in 4.5.1.

3.5 Proving Ground.- There shall be no break-up or separation of metal parts in the gun bore or in flight, when tested as specified in 4.5.8.

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3.6 Warhead Material Defects

3.6.1 Warhead Body.- The warhead body, including the cavity shall be free from cracks, splits, bursts, cold shuts, pipes, porosity, inclusions, folds seams and other metal defects.

3.6.2 Warhead Cavity.- In addition to the requirements of 3.6.1 the warhead cavity shall be free from scale, fins, burns, drawmarks, laminations, imbedded foreign matter, pits and all surface discontinuities which have a radius smaller than .030 inches.

3.6.3 Visual Standards.- Visual Standards for evaluation of metal defects will be established for each contract after the first months production. However, establishment of visual standards will not waive the requirements for compliance with 3.6.1 and 3.6.2 (see 6.8).

3.6.4 Insulator.- The insulator shall be free from cracks, seams, porosity, cavities and other defects.

3.7 Ultrasonic Inspection.- The warheads shall be free of any discontinuities equal or in excess of the inspection criteria within their respective zones when tested in accordance with 4.5.3. The inspection criteria are notches and holes in standards (see 9246484, 9246485 and 9246486) which simulate larger actual cracks and piping flaws."

3.8 Interior Cavity Finish

3.8.1 The interior cavity after painting shall be free from bare spots, blisters, grease, dirt and other foreign matter.

3.8.2 There shall be no pool of paint in the bottom of the cavity prior to palletizing.

3.9 Surface Finish.- The requirements for surface finish are as detailed on the applicable drawing. The roughness comparison specimens shall be used as a basis for surface roughness determination.

3.10 Eddy Current Inspection.- The warheads shall be free of any discontinuities equal to or in excess of the inspection criteria within their respective zones when tested in accordance with paragraph 4.5.9. The inspection criteria are notches machined into a standard (see 9246486) which simulate actual cracks. Rejection levels shall be set at 1/3 the height of the signal received from these notches.

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3.11 Slow Bend Precracked Charpy Test.- The nominal crack strength shall be greater than 25 KSI In. (Information only).

3.12 First Article Inspection.- This specification makes provisions for First Article Inspection. Requirements for the submission of First Article sample by the contractor shall be specified in the contract.

3.13 Workmanship.- The requirements for workmanship are as shown on the applicable drawing, in the applicable specifications and the following:

3.13.1 Surface Preparation.- Preparation of metal surfaces shall be as specified on the applicable drawings and in the applicable specifications referenced thereon. Where cleaning and preparation are required by component details, the finish specified shall apply to indicated surfaces regardless of contour configuration.

3.13.2 Painting.- Painting shall comply with the requirements of the applicable drawing. All paint shall be dry to the touch before packing for shipment. Drying time prior to testing shall be in accordance with the applicable specification.

3.13.3 Threads.- Threads shall be full and undamaged for the entire minimum length or depth as specified on the applicable drawing, and shall be in accordance with the provisions of specification MIL-A-2550.

3.13.4 Burr.- No part shall have a burr which might interfere with the assembly or function of the round or which might be injurious to personnel handling the item.

3.13.5 Foreign matter.- No part or assembly shall contain dirt, grease, chips, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions.- Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspections.- The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

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4.3 First Article Inspection

4.3.1 Submission.- The contractor shall submit a First Article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The First Article sample shall consist of the following items in sample quantities as indicated:

Part Description	Drawing	Quantity
Warhead Bonding Ass'y	9235997	25
Warhead (Unpainted Metal Parts)	9235995	25
Warhead (Prior to Knurling and Threading Base)	9235995	25
Warhead Insulation	9235996	10 Consecutively produced from each cavity of each mold
Warhead Bonding Ass'y (for Ballistic Testing, See 4.3.2.2.1).	9235997	20

All assemblies and components shall have been produced by the contractor or furnished by a supplier and shall have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample including statements of findings for materials, processes and tests. A sample containing known defects will not be submitted unless specifically authorized, by the Contracting Officer. A First Article sample, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.3.2 Inspections to be Performed.- See MIL-A-48078 and Table I specified herein.

4.3.2.1 Ballistic Tests

4.3.2.1.1 Submission.- Twenty complete assemblies packed in the form that will be used for shipping of the regular production quantity shall be submitted to a Government

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installation designated by the Contracting Officer for loading and ballistic testing. Unless otherwise directed by the Contracting Officer, the assemblies shall not be submitted prior to the acceptance of the sample submitted to the Government for examination (see 4.3.1) and shall consist of twenty assemblies.

4.3.2.1.2 Tests to be performed and acceptance criteria.- Proving ground tests will be performed on twenty (20) assemblies in accordance with 4.4.3.3.1, utilizing the procedures of 4.5.8 to determine compliance with 3.5.

4.3.3 Rejection.- See MIL-A-48078

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Warhead Bonding Assembly and Components				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Warhead Bonding Assy (Dwg. 9235997)				
	Examination for Defects	25	100%	3.2	4.4.2.4/4.4.2.7
	Bonding Assembly Pull Test	25	100%	3.2	4.5.8
	Warhead Unpainted Metal Parts (Dwg. 9235995)				
	Examination for Defects	25	100%	3.2	4.4.2.3
	Visual Inspection	25	100%	3.6	4.5.2
	Mechanical Properties Test	2	100%	3.3	4.5.4
	Radius of Base Cavity Test	2	100%	3.2	4.5.5
	Interior Surface Test	2	100%	3.2	4.5.6
	Warhead Insulation (Dwg. 9235996)				
	Examination for Defects	10/mold	100%	3.2	4.4.2.5
	Warhead (Prior to Knurling and Threading Base) (Dwg. 9235995)				
	Examination for Defects	25	100%	3.2	4.4.2.2
	Hydrostatic Test	25	100%	3.4	4.5.1
	Ultrasonic Test	25	100%	3.7	4.5.3
	Eddy Current Test	25	100%	3.10	4.5.9

NOTES:

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4.4 Quality Conformance Inspection

4.4.1 Inspection lot formation.- Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition for the first three lots produced, heat treatment inspection lots shall not exceed any of the following: One mill-heat of steel, eight hours production or 500 components. After three production lots, the one mill-heat restriction shall no longer apply. In any event, heat treatment inspection lots shall be restricted to components heat treated in the same furnace at the same temperature and for the same length of time. Test samples shall be selected after final heat treatment. Where heat treatment is performed at the same time in a batch-type furnace, additional identification control must be maintaned which directly relates to that portion of the inspection lot produced as a batch. The method of control must meet with approval of the Government. Components in heat treatment inspection lots shall remain grouped together for identifying purposes until mechanical property tests and interior surface and internal radius examinations have been completed. Components from heat treatment inspection lots which comply with applicable requirements may be subsequently re-grouped into final assembly lots.

4.4.2 Examination.- See MIL-A-48078.

a. Sampling Plans.- Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for Major and Minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-W-50577C (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Warhead, Prior to Nosing	1 OF 1	9235995
			NEXT HIGHER ASSEMBLY: 9235997

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Visual inspection		100%	3.6	4.5.2
Major	None defined				
Minor					
201	Evidence of poor workmanship		1.0%	3.13	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-W-50577C (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.2	Warhead, Prior to threading and knurling base		9235995
			NEXT HIGHER ASSEMBLY 9235997

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Hydrostatic test		100%	3.4	4.5.1
2	Hydrostatic test stamp missing		100%	3.2	Visual
3	Ultrasonic test		100%	3.7	4.5.3
4	Ultrasonic test stamp missing		100%	3.2	Visual
5	Eddy Current test		100%	3.10	4.5.9
6	Eddy Current test stamp missing		100%	3.2	Visual
Major	None defined				
Minor	None defined				

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-W-50577C (AR)

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.3	Warhead, Prior to knurling and painting		9235995
			NEXT HIGHER ASSEMBLY
			9235997

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Interior surface test	(b)		3.6	4.5.6
Major					
101	Major diameter of base thread (min.)		100%	3.2	Gage
102	Pitch diameter of base thread		0.25%	3.2	Gage
103	Pitch diameter of nose thread		0.40%	3.2	Gage
104	Minor diameter of nose thead		0.40%	3.2	Gage
105	Runout of pitch diameter of nose thread with bourrelet (in two locations), including seating of gage on shoulder		0.40%	3.2	Gage
106	Runout of datum at nose end with bourrelet		0.40%	3.2	Gage
107	Runout of datum in body with bourrelet		0.40%	3.2	Gage
108	Runout of pitch diameter of base threads with bourrelet (in two locations), including seating of gage on shoulder		0.40%	3.2	Gage
109	Diameter of bourrelet (minimum), (min.)		0.40%	3.2	Gage
110	Thickness of base, min.		0.40%	3.2	Gage
111	Wall thickness at rear datum		0.40%	3.2	Gage
112	Wall thickness of middle datum		0.40%	3.2	Gage
113	Wall thickness of Ogive (three places)		0.40%	3.2	Gage
114	Length of base thread including thread undercut		0.40%	3.2	Gage
115	Length of thread undercut,maximum(Max)		0.40%	3.2	Gage

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	CONTINUATION SHEET	2 OF 2	9235995
			NEXT HIGHER ASSEMBLY 9235997

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
116	Diameter of base thread undercut, min.		0.40%	3.2	Gage
117	Runout of base thread undercut with bourrelet		0.40%	3.2	Gage
118	Mechanical Properties Test				
	Tension	(a)		3.3	4.5.4
119	Radius of Base Cavity Test	(b)		3.2	4.5.5
120	Distance from nose to Datum at nose		0.40%	3.2	Gage
121	Distance from nose to Body Basic Diameter		0.40%	3.2	Gage
122	Precracked Charpy Test	(b)		3.11	4.5.11/Appendix
Minor					
201	Total length		0.65%	3.2	Gage
202	Diameter of recess in base		0.65%	3.2	Gage
203	Depth of recess in base		0.65%	3.2	Gage
204	Thickness of base, max.		0.65%	3.2	Gage
205	Distance from nose face to start of internal radius, max.		0.65%	3.2	Gage
206	Length of nose thread, min.		0.65%	3.2	Gage
207	Surface finish in base thread undercut		0.65%	3.2	Visual
208	Exterior surface finish		0.65%	3.9	Visual
209	Evidence of poor workmanship		1.0%	3.13	Visual

NOTES: (a) Special sampling plan (See 4.4.3.1).
(b) Use warheads selected per 4.4.3.1 for Major Defect 118.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-W-50577C (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.4	Warhead		9235995
			NEXT HIGHER ASSEMBLY 9235997

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Interior surface defects (After Phosphate)		100%	3.6	Visual
2	Pool of paint in bottom of cavity or blisters		100%	3.8.2	Visual
Major					
101	Diameter of bourrelet, max. (After Painting)		100%	3.2	Gage
102	Knurl missing, improper, or inadequate		100%	3.2	Visual
103	Interior protective coating inadequate including bare spots, grease, dirt, or other foreign matter.		100%	3.8.1	Visual
Minor					
201	Painted surface with total exposed or damaged area in excess of 1/4 sq. in.		0.65%	3.2	Visual
202	Marking missing, misleading or unidentifiable		0.65%	3.2	Visual
203	Compound missing from threads		0.65%	3.2	Visual
204	Evidence of poor workmanship		1.0%	3.13	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Insulation, Warhead	1 OF 1	9235996
			NEXT HIGHER ASSEMBLY 9235997

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Thickness of hub		0.40%	3.2	Gage
102	Diameter of hub		0.40%	3.2	Gage
103	Outside diameter		0.40%	3.2	Gage
Minor					
201	Thickness of step section		0.65%	3.2	Gage
202	Flatness		0.65%	3.2	Gage
203	Evidence of poor workmanship		1.0%	3.13	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-W-50577C (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.6	Warhead, Prior to Installing Insulation		9235997
			NEXT HIGHER ASSEMBLY 9235998

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Warhead base improperly finished		0.40%	3.2	Visual
102	Warhead base not clean or dry		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.0%	3.13	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-W-50577C (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.7	Warhead, After Assembly of Insulation		9235997
			NEXT HIGHER ASSEMBLY 9235998

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Proving Ground Test	(a)	-	3.5	4.5.8
Major					
101	Bonding Assembly Pull Test		100%	3.2	4.5.7
102	Insulation or adhesive above minimum recess		0.40%	3.2	Gage
Minor					
201	Lifting Plug Missing		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.0%	3.13	Visual

NOTES: (a) Special Sampling Plan (See 4.4.3.3).

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4.4.3 Testing

4.4.3.1 Mechanical Properties

4.4.3.1.1 Hardness.- Stations T and X (see dwg. 9235995)- Each final heat treated warhead from a heat treated lot shall be serialized and subjected to a hardness test on the surfaces indicated. Hardness readings shall be recorded for each serialized warhead. The test shall be conducted in accordance with American Society for Testing and Materials, Method E18. The average hardness reading thus obtained shall be used as the basis for selecting the tension test samples of paragraph 4.4.3.1.2.

4.4.3.1.2 Tension (see dwg. 9235995 and 4.4.2.3).- Two final heat treated warheads from a heat treated inspection lot shall be selected for the tension test. The warheads selected shall be those which have the highest average and the lowest average hardness reading from 4.4.3.1.1. From each of these warheads two specimens shall be tested in accordance with 4.5.4.

4.4.3.1.3 Retest.- In the event of rejection, new samples of higher or lower hardness values or both, as applicable, may be selected and tested in accordance with 4.4.3.1.2 in order to establish an acceptable hardness range. Once this range has been established, all warheads whose hardness values fall outside the range shall be rejected or subjected to retemper cycle if approved by the Government. An acceptable hardness range is one which will provide acceptable yield and elongation results.

4.4.3.2 Molded components.- The first ten (10) warhead insulations consecutively produced from each cavity of each mold shall be inspected for all drawing dimensions. In addition, ten (10) consecutive sample pieces shall be selected and inspected after a maximum of 5,000 pieces have been produced. If any part fails to meet all drawing requirements, production shall be stopped and the mold corrected. Deviations or suspected deviations in product shall be examined to determine if corrective action is required.

4.4.3.3 Proving Ground.- Security- Testing shall be performed as specified in 4.5.8.

4.4.3.3.1 For First Article or beginning with the first lot produced and continuing until three consecutive lots have complied with all of the applicable requirements specified, 20

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sample warheads shall be observed for security of parts when tested in accordance with 4.5.8. If any of the sample warheads fail to comply with the requirements for security of parts, the lot shall be rejected.

4.4.3.3.2 After three consecutive lots have complied with 4.4.3.3.1, 10 sample warheads shall be randomly selected from each lot for this test. The 10 sample warheads shall be observed for security of parts. If any of the sample warheads fail to comply with the requirement for security of parts, the lot shall be rejected.

4.4.3.4 Protective Coating - Salt Spray.- The sample size and acceptance criteria shall be as specified in the applicable specification. Failure to comply with the requirements shall be cause for taking the corrective action prescribed in TT-C-490.

4.4.3.4.1 Protective Coating - Thickness.- Two samples shall be selected at random to represent each four hour period, or less, of continuous paint production. The samples shall be inspected using the equipment in accordance with 4.5.10.2. The sample warheads shall be inspected for paint coating thickness at a minimum of four separate locations on each warhead. Any warhead failing to meet the applicable requirement shall be classed defective and every warhead painted since the last acceptable sample shall be rejected.

4.4.4 Inspection Equipment.- The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and paragraph 6.3 herein.

4.5 Test Methods and Procedures

4.5.1 Hydrostatic test.- The unpainted warhead shall be mounted in a suitable test fixture (see 4.4.4) and be subjected to the pressure specified on the applicable drawing. Any warhead which fails to comply with the applicable requirements shall be classed defective and removed from the lot.

4.5.2 Visual Examination.- The internal cavity of the warhead shall be inspected visually prior to nosing using peripheral lighting to assure conformance with the applicable requirements. Visual standards in accordance with 3.6.3 shall be set up for use during this inspection. Any warhead failing to meet the applicable requirements of 3.6 shall be classed defective and removed from the lot.

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4.5.3 Ultrasonic Inspection.- The warhead shall be ultrasonically inspected after hydrostatic testing but prior to threading the base using equipment and procedures in accordance with 4.5.3.1. The warhead shall be inspected with ultrasonics throughout all of the material, except in the region of threading and Eddy Current inspection. Ultrasonic energy shall be used at right angles to assure that orientation does not prevent detection of a defect. Three standards in accordance with 4.5.3.3 shall be used to check the effectiveness of the inspection process. Any warhead failing to meet the requirements of 3.7 shall be classed defective and removed from the lot.

4.5.3.1 Equipment and Procedures.- The ultrasonic inspection shall be performed in accordance with Dwg. 9225684 with the use of a liquid to assure a reliable coupling to transmit ultrasonic energy. Only immersion testing shall be used.

Using ultrasonic energy at right angles requires at least two separate sources of ultrasonic energy. This means that in the nose, ogive and bourrelet regions, ultrasonic energy shall move within the walls of the warhead with separate significant axial and circumferential components of velocity to detect flaws located anywhere on the outside, inside or within the walls. For the base region, significant ultrasonic velocity components shall separately be radial and circumferential.

Criteria for detection of cracks are notches which simulate actual larger cracks. Details are contained on Drawings 9246484, 9246485 and 9246486. Criteria for detection of piping in the base region consists of three drilled holes oriented parallel to the major axis of the warhead. During warhead rotation, the hole that comes second closest to the transducer is the target for locating the ultrasonic focal zone.

All notches and holes machined on the standards are targets for rejection. That is, any warhead that sends back an echo signal that is greater than or equal to the signal from the standard shall be rejected.

4.5.3.2 Calibration and Operation Procedures and Equipment.- Prior to initiation of ultrasonic inspection, the contractor shall submit for written approval one copy of his proposed procedures, including descriptions of equipment (contractor design) and operation procedures (see 6.3). No change to the procedure shall be made after the original approval without submission and approval of the proposed changes. As a minimum the adequacy of the inspection process shall be checked by using

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the standards at the start and finish of each production shift or change in operator, as well as at four (4) hour intervals during production. All channels shall properly indicate a rejection signal when encountering the notches and holes mentioned in 3.7. If the inspection process is determined to be inadequate, the condition shall be corrected and all items inspected since the last acceptable inspection check shall be reinspected.

Equipment used shall provide for a positive activation of a buzzer or alarm as well as a light system for each channel to alert the operator of the signal indications that are the criteria for rejection. A log book shall be maintained by the operator of each system which as a minimum, documents the use of standards, number of items inspected, number of rejects by zone, changing of transducers, instrumentation, repairs, reasons for equipment breakage, if any, and changing of qualified operators or inspectors. This log book shall be made available when requested for Government review.

4.5.3.3 Ultrasonic Standards.- Ultrasonic inspection standards shall be provided by the contractor for use during the ultrasonic inspection. These standards shall be machined in accordance with drawings 9246484, 9246485, and 9246486 and shall be inspected for all drawing requirements to determine their acceptability for use. Government verification of all inspections performed on the standards is required. In addition, a certificate of conformance for the standards shall be submitted to Commander, ARRADCOM, ATTN: DRDAR-QAR-I, Dover, New Jersey 07801, prior to use of the standards.

4.5.3.4 Operator Qualification.- Personnel performing the ultrasonic inspection shall be certified in accordance with MIL-STD-410 with the following exception:

"Each facility shall have as a minimum an individual who qualifies as a Level II inspector in all respects but need not include experience."

4.5.4 Mechanical Properties - Tension Testing.- (See 4.4.2.3). The two warheads selected for testing in accordance with 4.4.3.1.2 shall have two specimens taken from each and tested as follows:

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Base Section.- The base section tensile specimen shall be taken transversely, one inch from the centerline. Failure to meet the yield strength or elongation requirements of dwg. 9235995 shall require a retest from the remaining portion of the base results of the retest specimen and the first failing specimen shall be averaged to determine compliance with the applicable drawing requirement. Failure of the average results to meet the drawing requirement shall be cause for rejection of the heat treated inspection lot. In addition, failure of either individual test specimen to meet 130,000 psi minimum yield strength and 3% minimum elongation shall be cause for rejection of that heat treatment inspection lot represented by the sample.

Side Wall.- One specimen shall be taken from between stations T and X. Failure of any specimen to comply with the applicable drawing requirement shall be cause for rejection of the heat treated inspection lot.

Testing shall be conducted in accordance with American Society for Testing and Materials, Method E8 with the largest substandard size round specimen possible. The contractor shall submit a copy of all test reports monthly in accordance with Data Item Description DI-T-1906 reporting all information required.

4.5.5 Radius of Base Cavity (See dwg. 9235995).- The warheads use in 4.4.3.1.2 shall be gaged for proper radius when the warheads are cut for tension specimens. If any warhead fails to comply, the heat treated inspection lot shall be rejected.

4.5.6 Interior surface (See 3.6).- The warheads used in 4.4.3.1.2 shall be visually examined for interior surface discontinuities when the warheads are cut for tension specimens. Failure to meet the requirements of 3.6 shall be cause for rejection of the heat treated inspection lot.

4.5.7 Bonding Assembly Pull Test (See dwg. 9235997).- This test shall be performed 100 percent, using equipment in accordance with 4.4.4. A tensile load in accordance with Dwg. 9235997 shall be applied and any assembly which fails shall be classed defective and removed from the lot.

4.5.8 Proving Ground.- These tests shall be performed at a Government Proving Ground in accordance with the applicable Acceptance Test Procedure. The sample warheads selected in accordance with 4.4.3.3 shall be loaded and assembled as specified on dwg. 9236005 and fired at ambient temperature in a M185 or M199 series cannon using a proof propelling charge which produces an average piezo equivalent pressure of 54,500 psi to 58,800 psi chamber pressure based on copper pressure gages. The

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propelling charges shall be uniformly temperature conditioned prior to the test. Pressure-time traces and differential pressures shall be obtained for information only for each round fired by means of a spindle tap in the breech of the cannon and a piezo gage at the projectile base location. Adequate photographic and instrumental coverage shall be provided to assure compliance with 3.5. Range and deflections shall be recorded for informational purposes only. In the event of any abnormal condition, as observed by personnel or instrumentation, an attempt shall be made to recover and examine the metal parts for informational purposes only.

4.5.9 Eddy Current Inspection.- The warhead shall be Eddy Current inspected after Hydrostatic and Ultrasonic inspection using equipment and procedures in accordance with 4.5.9.1. The warhead shall be inspected in the zones established on a standard in accordance with 4.5.9.2. Any warhead failing to meet the requirements of 3.10 shall be classed defective and removed from the lot.

4.5.9.1 Calibration and Operation Procedures and Equipment.- Prior to initiation of Eddy Current inspection, the contractor shall submit for written approval one copy of his proposed procedures, including descriptions of equipment (contractor design) and operation procedures (see 6.3). No change to the procedure shall be made after the original approval without submission and approval of the proposed changes. As a minimum the adequacy of the inspection process shall be checked by using the standards at the start and finish of each production shift or change in operator, as well as at four (4) hour intervals during production. All channels shall properly indicate a rejection signal when adjusted to the notches mentioned in 3.12. If the inspection process is determined to be inadequate, the condition shall be corrected and all items inspected since the last acceptable inspection check shall be reinspected. A log book shall be maintained by the operator of each system which as a minimum documents the use of standards, number of items inspected, number of rejects by zone, changing of probes, instrumentation, repairs, reasons for equipment breakage, if any, and changing of qualified operators or inspectors. This log book shall be made available when requested for Government review.

4.5.9.2 Eddy Current Standard.- An Eddy Current inspection standard shall be provided by the contractor for use during the Eddy Current inspection. This standard shall be machined in accordance with Drawing 9246486 and shall be inspected for all drawing requirements, to determine its acceptability for use. Government verification of all inspections performed on the standard is required. In addition, a certificate of conformance for the standard shall be submitted to Commander, ARRADCOM, ATTN: DRDAR-QAR-I, Dover, New Jersey 07801, prior to use of the standard.

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4.5.9.3 Operator Qualification.- Personnel performing the Eddy Current inspection shall be certified in accordance with MIL-STD-410 with the following exception:

"Each facility shall have as a minimum an individual who qualifies as a Level II inspector in all respects but need not include experience."

4.5.10 Protective Coating.- The procedures for testing the cleaning of components and the application of protective coating shall be in accordance with the requirements of the applicable drawing and the specification listed thereon.

4.5.10.1 Salt Spray Testing.- Salt spray testing shall be accomplished on production projectiles or sectioned projectiles that were acceptable at the time of painting and were sectioned subsequent to painting. If sectioned projectiles are used, sectioning shall be accomplished to allow testing of all major areas of the projectile on an alternating basis. The salt spray test shall be performed utilizing the equipment and procedures specified by Federal Test Method Standard No. 141 and TT-C-490.

4.5.10.2 Protective Coating - Thickness.- Protective coating thickness shall be accomplished using commercially available equipment.

4.5.11 Slow Bend Precrack Charpy Test

4.5.11.1 Initial Production Testing.- Beginning with the initial lot, the warheads selected for testing in accordance with 4.4.3.1 for high hardness shall have eight specimens taken and tested as follows:

Base Section.- Four specimens shall be taken as detailed in Figure 1. Samples number one and two shall be tested as minus sixty-five degrees fahrenheit (-65°F) and samples three and four shall be tested at seventy degrees fahrenheit (+70°F). The notches shall be machined across the rear face of the specimen.

Sidewall.- Four specimens shall be selected longitudinally from between stations T and X of Drawing 9235995. Two (2) samples each shall be tested at minus sixty-five degrees fahrenheit (-65°F) and seventy degrees fahrenheit (70°F). The notches shall be machined on the outside surface of the specimen.

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Testing shall be conducted in accordance with the test procedure defined in the Appendix of this specification. The contractor shall submit a copy of all test reports monthly in accordance with Data Item Description DI-T-1906 reporting all information required by the Appendix.

4.5.11.2 After 100 heat treat lots have been produced the sample size shall be reduced to four samples per heat treat. Only the minus sixty-five degree fahrenheit (-65°F) test shall be performed.

Testing shall be conducted in accordance with the test procedure defined in the Appendix of this specification. The contractor shall submit a copy of all test reports monthly in accordance with Data Item Description DI-T-1906 reporting all information required by the Appendix.

5. PREPARATION FOR DELIVERY

5.1 Packing - Level C.- The warhead bonding assembly shall be packed in accordance with dwg. 9327883.

5.2 Marking.- Marking shall be in accordance with Dwg. 9327883. Insert descriptive nomenclature as follows:

8 Warhead, Bonding Assembly, 155MM, M549 RAP
LOT Number

6. NOTES

6.1 Intended Use.- The components covered by this specification are intended for use on the 155MM, M549/M549A1, RA Projectile.

6.2 Ordering Data.- See MIL-A-48078 (PA)

6.3 Submission of Inspection Procedures and/or Equipment Designs for Approval.- See MIL-A-48078. Submit equipment designs, as required, to Commander, ARRADCOM, ATTN: DRDAR-QAR-I, Dover, New Jersey 07801.

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6.4 Lot Sizes.- It is noted that the size of inspection lots of components, assemblies or items of delivery may differ from the actual quantities contractually scheduled for delivery. However, in order to facilitate scheduling of tests or deliveries, inspection lots of items of delivery may be equivalent to such contract quantities provided all of the lot formation criteria and sampling provisions of this specification are maintained.

6.5 Visual examination qualification.- When compliance with the applicable requirement is in doubt as a result of visual examination the characteristic may be measured or gaged to determine acceptability.

6.6 Proving Ground Test Summary

Test	Sample Size (see 4.4.3.3)		Requirements
Security	* 20	10	No break-up or metal parts separation (see 3.5)

* Applicable only for First Article Test and first three lots.

6.7 Submission of ballistic test data and ammunition data cards.- In addition to the normal distribution of records for items procured by the Department of the Army, two copies of all ballistic test data and ammunition data cards shall be forwarded to Commanding Officer, ARRADCOM, Dover, New Jersey 07801, ATTN: DRDAR-QAR-Q and DRDAR-LCU-CA.

6.8 Visual Standards.- Visual standards to be established at contractors plant in compliance with 3.6.3.

6.9 Data Requirements.- Deliverable data required by this specification is cited in the following paragraph:

Paragraph	Data Requirement	Applicable DID
4.5.4	Test Report	DI-T-1906
4.5.11	Test Report	DI-T-1906

(Copies of data item descriptions required by the contractors in conjunction with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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6.10 The following notes are provided for information only.

6.10.1 Heat Treat - Austenitize - 1475° to 1550°F for 1 hour minimum.

- Oil Quench - 110°F to 140°F

- Temper - 1040°F min. for 1 hr. min.

6.10.2 Cooling of forgings should not be retarded. Slow cooling can cause the formation of carbides.

6.10.3 Spheroidizing after forging can be beneficial in rough machining, nosing and heat treatment.

6.10.4 Nosing should be performed in a minimum of two step.

6.10.5 The "As Forged" base thickness should be .9" maximum to assure adequate working of the base.

6.10.6 Quenching over spuds is recommended to assure adequate quenching of the base.

6.10.7 The Hydrotest plug should be designed with a one quarter inch, minimum thread relief to prevent loading of initial threads on the warhead.

6.10.8 Examination should be made of all hydrotest plugs after a warhead is blown during testing. Badly nicked or damaged plugs may cause destruction of acceptable items

6.11 Changes from Previous Issue.- Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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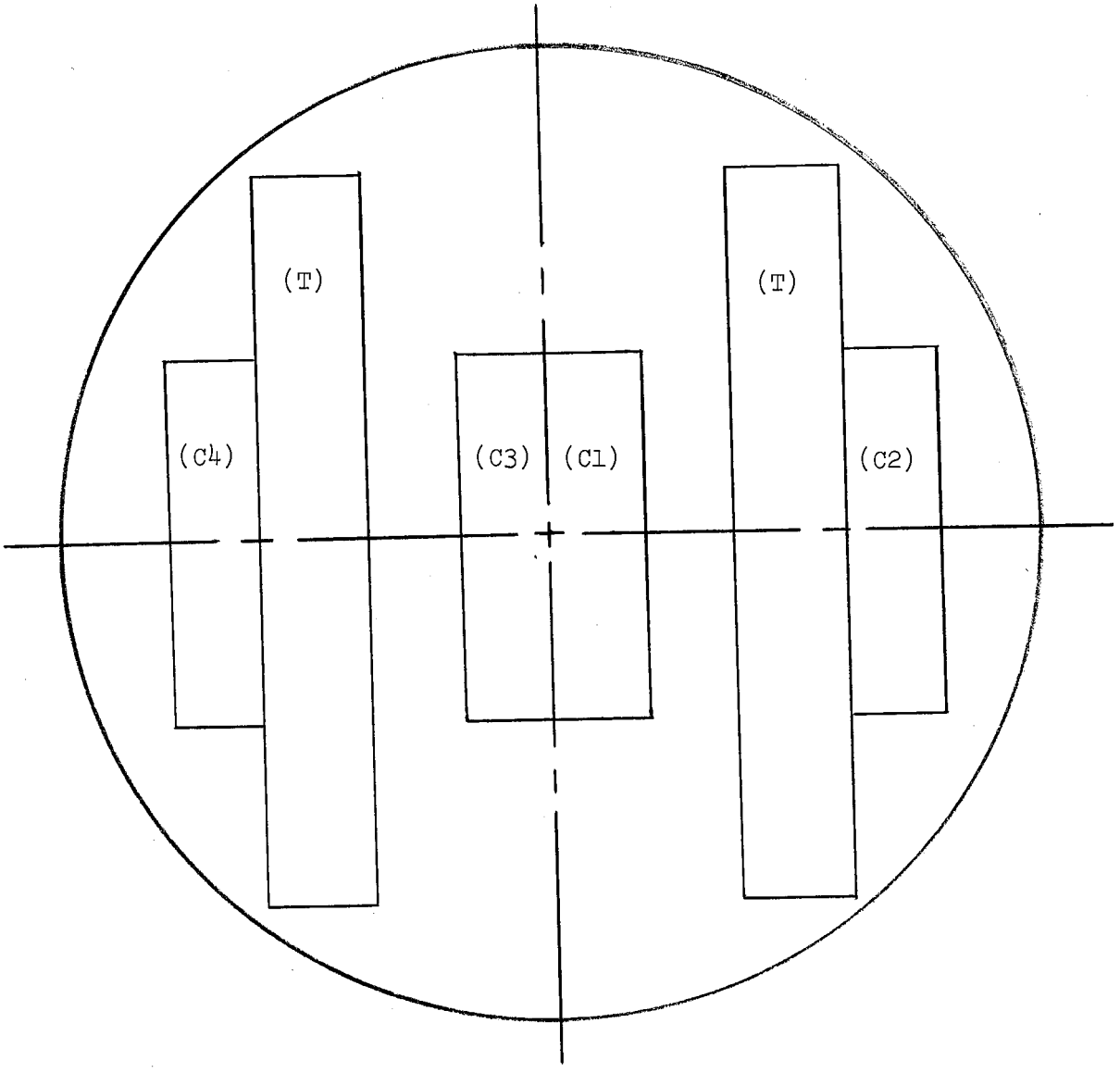


FIGURE 1

Approximate Location of Tension and Charpy Base Specimens

(T - Tensile, C- Charpy)

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APPENDIX

TEST METHOD FOR SLOW BEND PRECRACKED CHARPY TEST

10. SCOPE

10.1 Scope.- This appendix covers the test method used in determining the nominal strength of a Charpy specimen containing a fatigue crack and tested in slow bending.

20. APPLICABLE DOCUMENTS

20.1 - ASTM Standards

E 4 Verification of Testing Machines

E 139 Recommended Practice for Conducting Creep, Creep-Rupture and Stress Rupture Tests of Metallic Materials

E 399 Plane Strain Fracture Toughness of Metallic Materials

30. SUMMARY OF METHOD

30.1 - The method involves three point bend testing of notched and fatigue cracked specimens using fixtures which minimize the contribution of friction forces to the measured applied load. The maximum load in the test is recorded and the nominal crack strength is determined from this value and the original dimensions of the specimen using the simple beam equation.

40. PRECAUTIONARY NOTES

40.1 Precautionary Notes.- When preparing specimens and testing for nominal crack strength, it is important that certain precautions be kept in mind.

40.1.1 Crack Strength.- The crack strength of some materials will be sensitive to the rate of loading and loading rates should be confined to the range given in 80.3.2.

40.1.2 Temperature.- The crack strength will be influenced by the testing temperature and calibration relations may change with testing temperature.

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50. DEFINITIONS

50.1 Nominal Crack Strength, $\bar{\sigma}_C$.- As determined by this method, the nominal crack strength is defined as the nominal stress in a notched and cracked bend specimen calculated on the basis of the length of the initial uncracked ligament and the maximum load sustained in the test. The crack strength is analogous to the ultimate tensile strength in that it is based on the dimensions of the specimen before testing.

50.2 The following symbols are used in this test method.

a. - fatigue crack depth (In)

B - specimen thickness (In)

W - specimen depth (In)

M - Bending Moment (In - lb_f)

f(a/W) - Power series of ratio (a/W) from ASTM-E-399

$\bar{\sigma}_c$ - Nominal Crack Strength - (KSI)

P max - Maximum Load Applied - (lb_f)

S - Span Between Support Rollers - (In)

Kf(max) - Maximum Stress Intensity Factor (KSI $\sqrt{In}$)

K - Stress Intensity factor (KSI $\sqrt{In}$)

60. APPARATUS

60.1 Tension Testing Machine.- Tension-Testing Machine conforming to the requirements of ASTM E 4.

60.2 Bend-Test Fixture.- The procedure involves the testing of the specimen in three point bending. The bend fixture should be so designed as to minimize errors that arise from friction between the specimen and the supports. This can be accomplished by allowing the support rollers to rotate and move apart slightly during loading as described in ASTM E 399 (Fig. 1). The nominal span shall be 1.97 in. If the diameter of the support rollers is between 0.30 in. to 0.40 in. there will be a negligible change in the span during deformation of the specimen.

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60.3 Temperature Control.- The testing temperatures shall be controlled by use of an integral temperature conditioning chamber as part of the testing apparatus. Any suitable method may be used to temperature condition the specimen providing the region including the notched and cracked section can be uniformly maintained within $\pm 2^{\circ}F$ of the desired test temperature.

60.4 Temperature Measurement.- The temperature of the specimen during testing shall be measured by means of two thermocouples attached on either side of the notched and cracked section. These thermocouples may be spot welded in place, however, the spot weld locations should avoid the crack path. At the test temperature, the difference between the temperatures indicated by these two thermocouples should not exceed $1^{\circ}F$ and the average of these two temperatures shall be within $\pm 2^{\circ}F$ of the desired test temperature.

60.4.1 Thermocouples.- Temperatures shall be measured with calibrated thermocouples used in conjuntion with potentiometers or millivoltmeters. Such measurements are subject to various errors; reference should be made to ASTM E 139 for a discussion of errors.

60.4.2 Calibration.- The temperature measuring apparatus should be calibrated periodically against standards traceable to the National Bureau of Standards in order to ensure that accuracy specified above can be achieved.

70. TEST SPECIMENS, DIMENSIONS AND PREPARATION

70.1 Dimensions.- The precracked Charpy bend specimen is shown in Figure 1. The crack depth (including notch), a, is nominally equal to one-half the thickness, B, and shall lie within the range between 0.45 to 0.55 B. $\frac{}{2}$

70.1.1 Crack Starter.- The crack starter may consist of a "V" notch having an included angle between 45° and 60° provided with a small radius at its tip. Experience has shown that a notch radius of less than 0.003 inches is necessary in order to facilitate the initiation of a fatigue crack at a low stress intensity level.

70.2 Fatigue Cracking.- In order to ensure a sufficiently sharp fatigue crack and to facilitate the production of this crack, the following procedure shall be followed.

70.2.1 Specimen Condition.- The fatigue cracking shall be conducted with the specimen in the condition of heat treatment in which it is to be tested.

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70.2.2 Cracking Method.- The method used for fatigue cracking shall be such that the crack extension direction is perpendicular to the notched face of the specimen. Cantilever bending or three point bending may be used to produce fatigue cracks. Cantilever bending facilitates completely reversed loading which aids in starting and propagating the crack. For practical purposes, the K calibration for three point bending may be used to estimate the stress intensity factors.

NOTE 1 - The expression given in ASTM E 399 for the K calibration in three point bending may be rewritten in terms of the bending moment M as follows and used to compute the stress intensity in cantilever bending: $K = (4M/BW^{3/2})\ f(a/W)$ where the values of the function f(a/W) are given in tabular form in ASTM E 399.

NOTE 2 - The stress distribution at the crack tip in cantilever bending can be influenced by excessive clamping forces to the extent that the fatigue crack will deviate from a plane normal to the notched face of the specimen.

70.2.3 Cracking.- During the last 0.020 inches of fatigue cracking, the maximum intensity of the fatigue cycle, K_f (max) shall not exceed $20KpI^{-3/2}$. The stress intensity range should not be less than $0.9K_f$ (max).

80. TEST PROCEDURE

80.1 Number of Tests.- Samples shall be selected as specified in the basic specification.

80.2 Specimen Measurement.- All specimen dimensions shall be within the tolerances shown in Figure 1.

80.2.1 - Measure the thickness, B, to the nearest 0.001 inches at not less than two positions between the fatigue crack tip and the unnotched edge of the specimen and record the average positions, one on each side of the crack starter notch and record the average value.

80.2.2 - After fracture measure the initial fatigue crack depth, a, to the nearest 0.5 percent at the following three positions: at the center of the crack front and midway between the center and the inter-section of the crack front with the specimen surfaces. Use the average of these three measurements as the crack depth in the calculation of σ_c (see 90.1). If the difference between any two of the crack length measurements exceeds 5 percent of the average, or if part of the crack front

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is closer to the machined notch root than 5 percent of the average, the specimen should be discarded. Also, if the length of either surface trace of the crack is less than 80 percent of the average crack length, as defined above, the specimen should be discarded.

80.3 Bend Testing.- Set up the bend test fixture so that the line of action of the applied load shall pass midway between the support roll centers within 0.025 inches of the distance between these centers. Measure the span (distance between the support roll centers) within 1 percent and record this value as S. This value should fall between 1.9 inches and 2.1 inches. If not, the span should be adjusted to fall within this range.

80.3.1 - Locate the specimen with the crack tip midway between the roll centers to within 0.025 inches, and square to the roll axes within 2 degrees. Check the parallelism between the back face of the specimen and the loading roller by sighting the gap between these surfaces just before contact is made. As a small load is applied light in this gap should extinguish simultaneously across the thickness of the specimen. If this condition is not met, the loading roller should be suitably adjusted.

80.3.2 - Load the specimen at a rate between 700 lbf/min and 3600 lbf/min until the maximum load is reached and record this value as P_{max} (lbf).

NOTE 3 - It is desirable to follow the load change carefully because specimens having low toughness will break abruptly and no slow decrease in load beyond the maximum will be observed.

90. CALCULATION AND INTERPRETATION OF RESULTS

90.1 Calculations.- Calculate the nominal crack strength of the specimen in KSI from the measured value of P_{max} and the initial dimensions of the specimen as follows:

$$\sigma_C = 3000\ P_{max} S/2\ B(W - a)^2$$

90.2 Fracture Appearance.- The appearance of the fracture may furnish valuable supplementary information. A means for describing the fracture appearance of cracked bend specimens is given in ASTM E 399 and may be used to characterize the fracture appearance of the precracked Charpy specimens.

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100. REPORT

100.1 - The report shall include the following for each specimen tested:

100.1.1 - All specimen dimensions measured.

100.1.2 - The maximum load P_{max}

100.1.3 - The nominal crack strength

100.1.4 - Test temperature

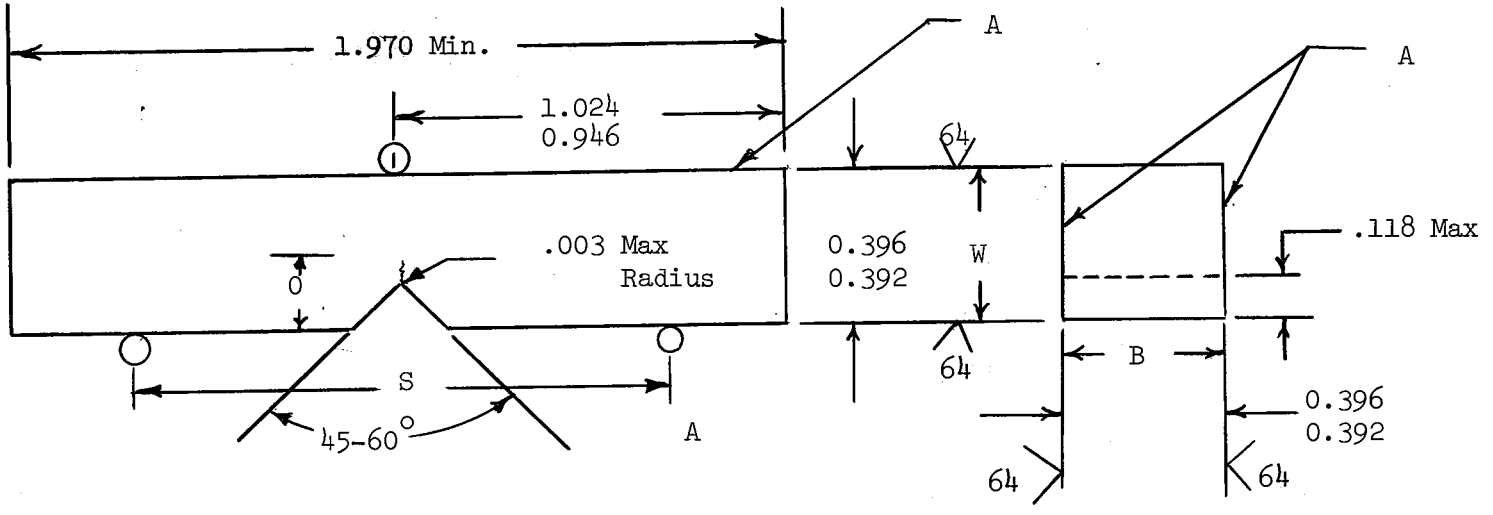
100.1.5 - The maximum stress intensity $K_f(max)$ during the final 0.020 inches of fatigue cracking.

100.1.6 - Crack plane orientation

100.1.7 - Fracture appearance

100.1.8 - 0.2 percent offset tensile yield strength

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NOTE 1. All dimensions in inches

NOTE 2. "A" surfaces shall be perpendicular or parallel as applicable to within 0.001 In TIR

NOTE 3. Crack starter shall be perpendicular to specimen length and thickness to within ±2 degrees.

FIGURE 1 Dimensions Of Precracked Charpy Specimens

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